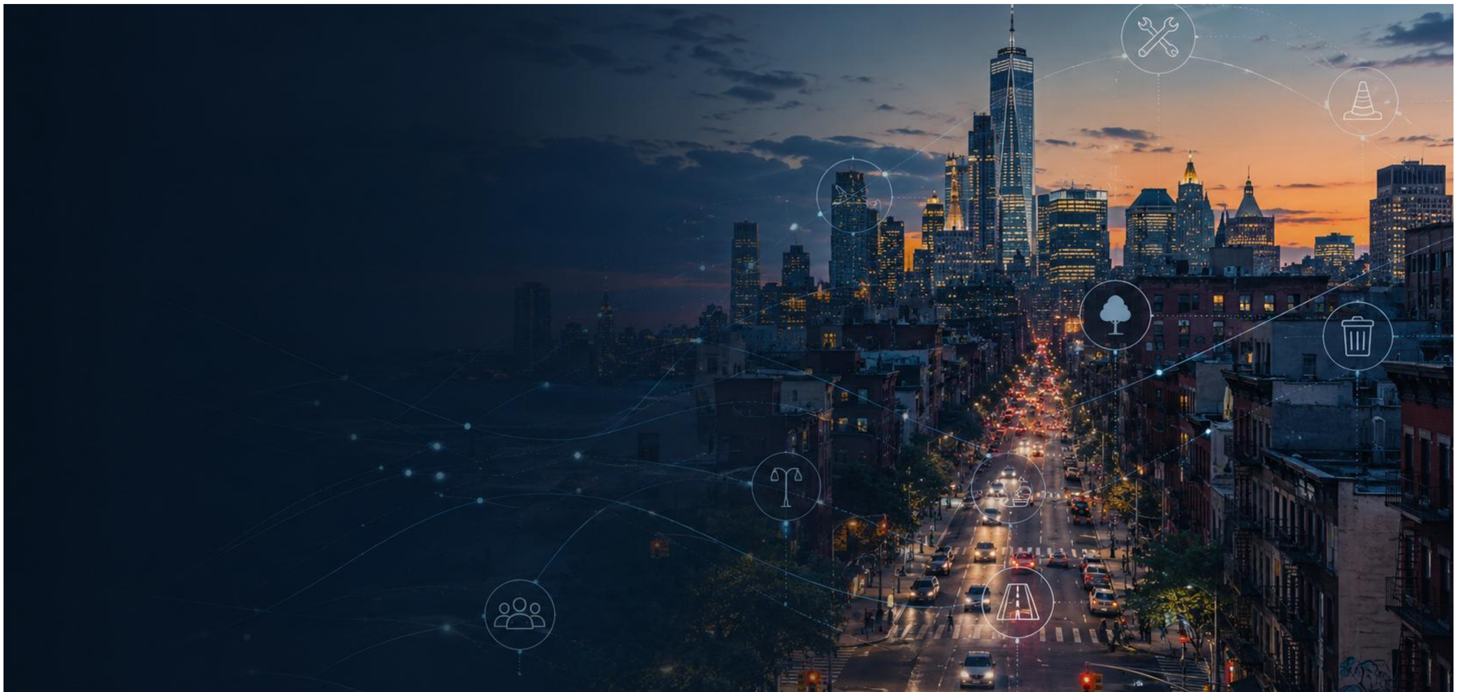

ITAI 3375 - CURATING DATA FOR AI OPTIMIZATION

NYC 311 Data Curation Issues Encountered and Technical Review

Curating NYC 311 Service Requests for AI-Based Urgency Classification

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Project Overview

By outlining the primary technical difficulties found during the curating of the NYC 311 Service Requests dataset and describing how the updated Python and Google Colab methodology resolved them, this document complements the NYC 311 final presentation.

Dataset	AI Use Case	Technical Environment
NYC 311 Service Requests from 2020 to Present	Preliminary classification of requests as High, Medium, Low, or Needs Review	Environment set up: Google Colab, Python, Pandas, NumPy, Matplotlib, Requests, and the NYC Open Data API

Issues Encountered

The duplicate and cleanliness evidence was not sufficiently obvious in the initial study. Each result became measurable, reviewable, and appropriate for the final presentation with the following changes.

1 Duplicate results were not visible

Issue: Although duplication checks were carried out in the first notebook, it was unclear from the results if exact duplicates, repeated service-request IDs, or potential repeated complaints were discovered.

Resolution: The revised notebook presents three separate checks: exact duplicate rows, duplicate unique_key values, and potential duplicate records based on date, agency, complaint type, descriptor, ZIP code, and borough.

2 Cleanliness changes were unclear

Issue: The previous result gave the impression that nothing had changed because it was difficult to compare missing data with existing values before and after curation.

Resolution: Missing counts, missing percentages, present-value counts, cleanliness percentages, and an overall cleanliness score both before and after cleaning are now displayed in a column-level cleanliness report.

3 A zero duplicate count needed explanation

Issue: Because unique_key is intended to identify a single service request, some downloaded samples may contain no exact duplicate IDs.

Resolution: Instead of generating or supposing duplicate records, the updated analysis displays any duplicate complaints independently and accepts zero as a valid result.

4 Some missing values were legitimate

Issue: Because **a request is still open or no deadline has been set, fields like closed_date, due_date, and resolution_description may be** empty**.**

Resolution: The workflow preserves legitimate missingness, uses transparent labels only for selected descriptive fields, and avoids inventing dates or operational outcomes.

5 Location data created a privacy trade-off

Issue: Full ZIP codes are useful for geographic analysis but increase the risk of identifying or profiling small groups when combined with other fields.

Resolution: The curating procedure eliminates the whole ZIP field from the curated output, generalizes ZIP codes to the first three digits, and preserves borough-level information.

Technical Review

The revised workflow is reproducible and audit-ready. Each stage produces visible evidence that can be used in the presentation, Data Card, and Q&A discussion.

Tools and Data Access

- • Google Colab and Python offer a notebook-based environment that can be replicated.
- • Data transformation, text normalization, duplicate detection, missing-data analysis, and profiling are all supported by Pandas and NumPy.
- • Matplotlib: creates before-and-after charts suitable for presentations.
- • Requests with Socrata API: use pagination to obtain a specific number of rows from NYC Open Data dataset ID erm2-nwe9.
- • The twelve project fields that were chosen include request IDs, dates, agency, complaint specifics, location, status, and resolution details.

Curation Workflow

1. Retrieve	2. Profile	3. Review duplicates	4. Clean & standardize	5. Protect location	6. Compare results	7. Export & document
Access and License:	use is governed by the NYC.gov NYC Open Data – 311 Service Requests from 2020 to Present.					
Availability	Free and publicly accessible					
Source:	NYC Open Data / NYC 311					
Usage Terms:	NYC.gov Terms of Use					

Main Technical Operations

Operation	Purpose	Evidence Produced
Standardize missing markers	Convert blank strings and placeholders into consistent missing values	Missing-value report before and after
Normalize text	Trim spaces and collapse repeated whitespace	Cleaning audit log and curated preview
Detect duplicates	Check exact rows, repeated service IDs, and potential repeated complaints	Duplicate summary and example tables

Operation	Purpose	Evidence Produced
Standardize categories	Canonicalize agency, borough, and status values	Consistency improvements in curated data
Validate dates	Parse dates and flag impossible closing chronology	Count of corrected or missing date values
Generalize ZIP codes	Reduce privacy risk while preserving broad geographic context	Curated data containing three-digit ZIP groups
Export outputs	Make the workflow reproducible and presentation-ready	CSV tables, charts, audit log, and cleaned dataset

Technical judgment: Duplicate counts and missing data should be interpreted rather than altered. Zero is a valid outcome, and valid missing values should be recorded instead of being filled in with made-up data.

Evidence Produced and Key Takeaways

The corrected code generates separate files for review, presentation, and documentation. These outputs directly support the pipeline, quality, governance, and reflection sections of the final project.

Output File	How It Supports the Project
raw_data.csv	Preserves the unmodified input subset for reproducibility.
cleaned_data.csv	Contains standardized fields and privacy-aware location handling.
duplicates_before.csv / duplicates_after.csv	Shows the duplicate checks before and after curation.
cleanliness_before.csv / cleanliness_after.csv	Reports missing and present values for every field.
cleanliness_comparison.csv	Provides the before-and-after evidence needed for the quality narrative.
cleaning_audit_log.csv	Documents each cleaning action, row counts, and technical reason.
cleanliness_before_after.png	Provides a readable chart for the Quality Results section.
duplicates_before_after.png	Visualizes duplicate findings before and after cleaning.

Key Takeaways

- • Visibility is important: Unless the outcome is clearly displayed in a table, chart, or audit log, a cleaning step is not persuasive. • Zero is still proof: The chosen sample's unique service-request identifiers can be shown by a zero duplicate count.
- • Missing does not always imply incorrect: There may be valid missing dates or resolutions for outstanding requests in operational datasets.
- • Technical design is impacted by governance: generalizing location data is a step in the curation process rather than an afterthought. • Human review is still required; rather than taking the role of agency judgment, preliminary urgency labels and possible duplicate flags should support review.

Final takeaway: Successful data curation requires more than running code. Every decision must be visible, measurable, documented, and connected to the intended AI use case.

Supporting code: NYC311_Duplicates_Cleanliness_Colab.ipynb and NYC311_Duplicates_Cleanliness_Code.py